organoid

sc

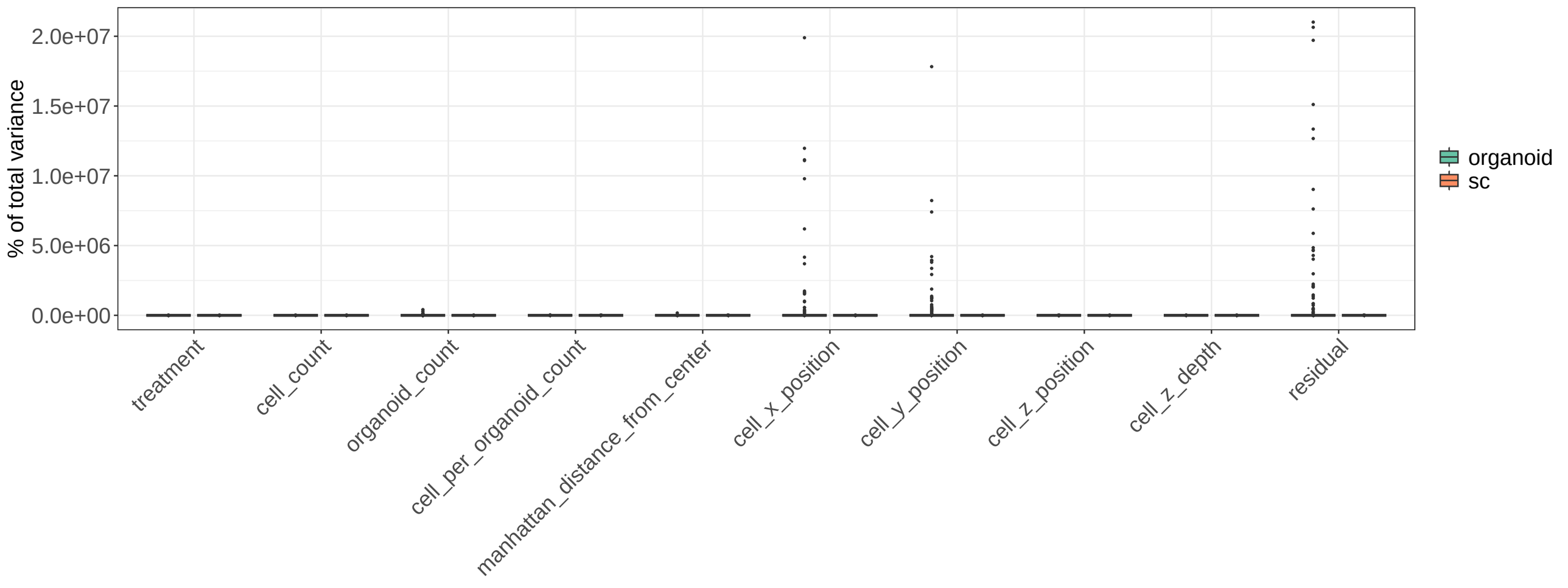
organoid_agg

sc_agg

−log10 FDR

treatment coefficient

R2


% of total variance
2.0e+07
1.5e+07
1.0e+07
5.0e+06
0.0e+00
treatment
cell_count
organoid_count
cell_per_organoid_count
manhattan_distance_from_center
cell_x_position
cell_y_position
cell_z_position
cell_z_depth
residual
organoid
sc

NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1

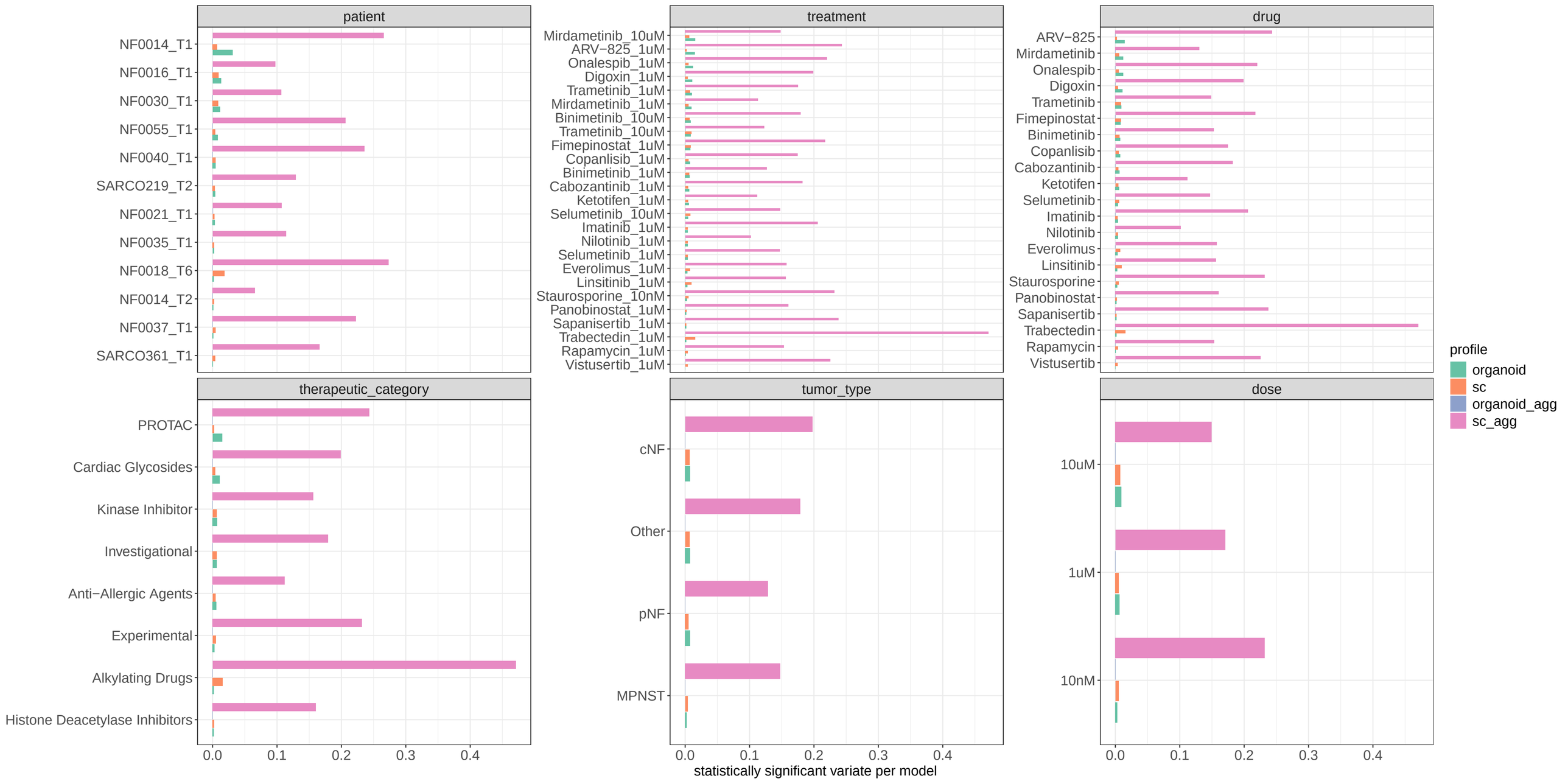
Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM

median treatment % var
4
3
2
1
0

NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1

Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM

median treatment % var
4
3
2
1
0

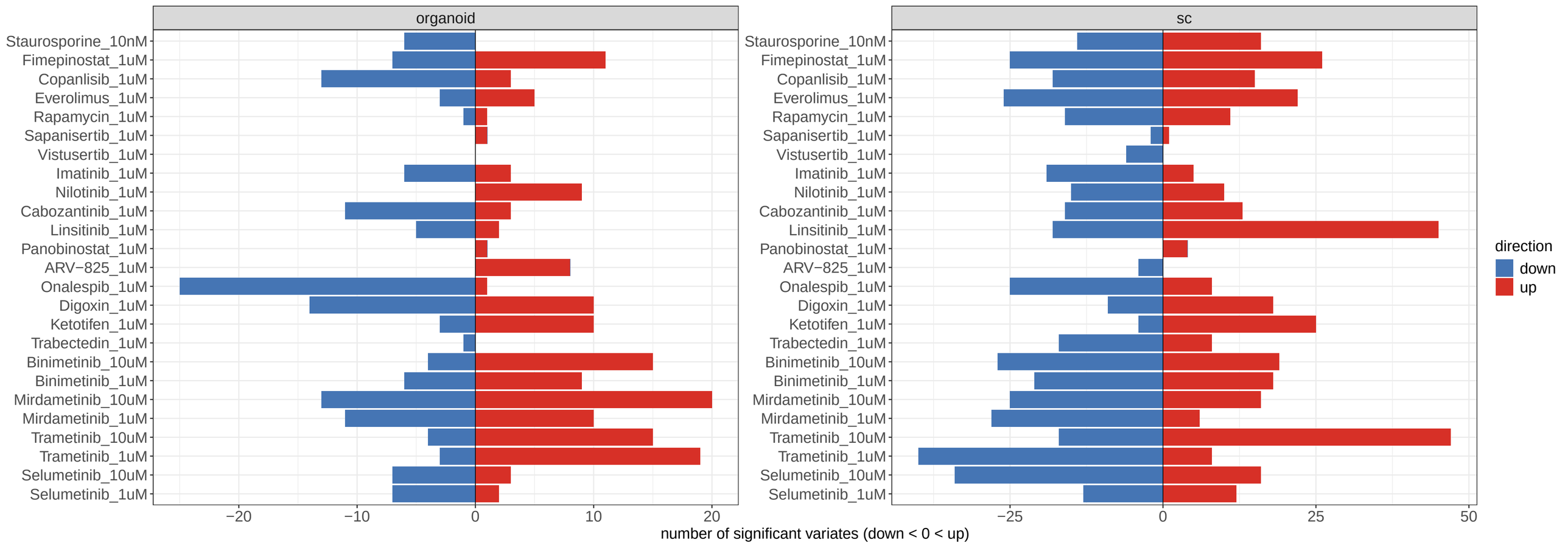

patient
NF0014_T1
NF0016_T1
NF0030_T1
NF0055_T1
NF0040_T1
SARCO219_T2
NF0021_T1
NF0035_T1
NF0018_T6
NF0014_T2
NF0037_T1
SARCO361_T1
treatment
Mirdametinib_10uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Trametinib_1uM
Mirdametinib_1uM
Binimetinib_10uM
Trametinib_10uM
Fimepinostat_1uM
Copanlisib_1uM
Binimetinib_1uM
Cabozantinib_1uM
Ketotifen_1uM
Selumetinib_10uM
Imatinib_1uM
Nilotinib_1uM
Selumetinib_1uM
Everolimus_1uM
Linsitinib_1uM
Staurosporine_10nM
Panobinostat_1uM
Sapanisertib_1uM
Trabectedin_1uM
Rapamycin_1uM
Vistusertib_1uM
drug
ARV−825
Mirdametinib
Onalespib
Digoxin
Trametinib
Fimepinostat
Binimetinib
Copanlisib
Cabozantinib
Ketotifen
Selumetinib
Imatinib
Nilotinib
Everolimus
Linsitinib
Staurosporine
Panobinostat
Sapanisertib
Trabectedin
Rapamycin
Vistusertib
profile
organoid
sc
organoid_agg
sc_agg
therapeutic_category
PROTAC
Cardiac Glycosides
Kinase Inhibitor
Investigational
Anti−Allergic Agents
Experimental
Alkylating Drugs
Histone Deacetylase Inhibitors
tumor_type
cNF
Other
pNF
MPNST
dose
10uM
1uM
10nM
0.0
0.1
0.2
0.3
0.4
statistically significant variate per model

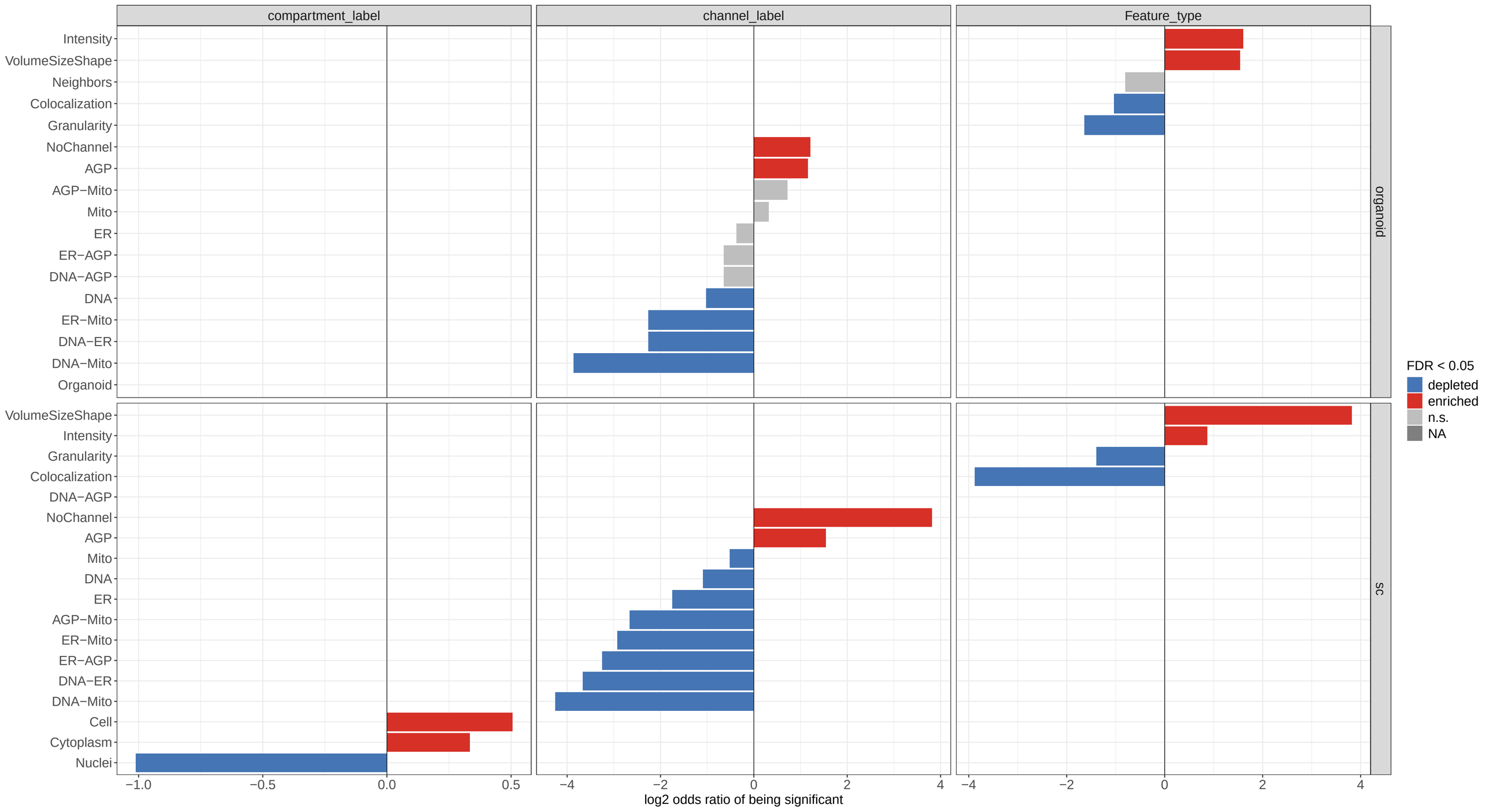
NF0014_T1
NF0014_T2
NF0016_T1
NF0018_T6
NF0021_T1
NF0030_T1
NF0035_T1
NF0037_T1
NF0040_T1
NF0055_T1
SARCO219_T2
SARCO361_T1

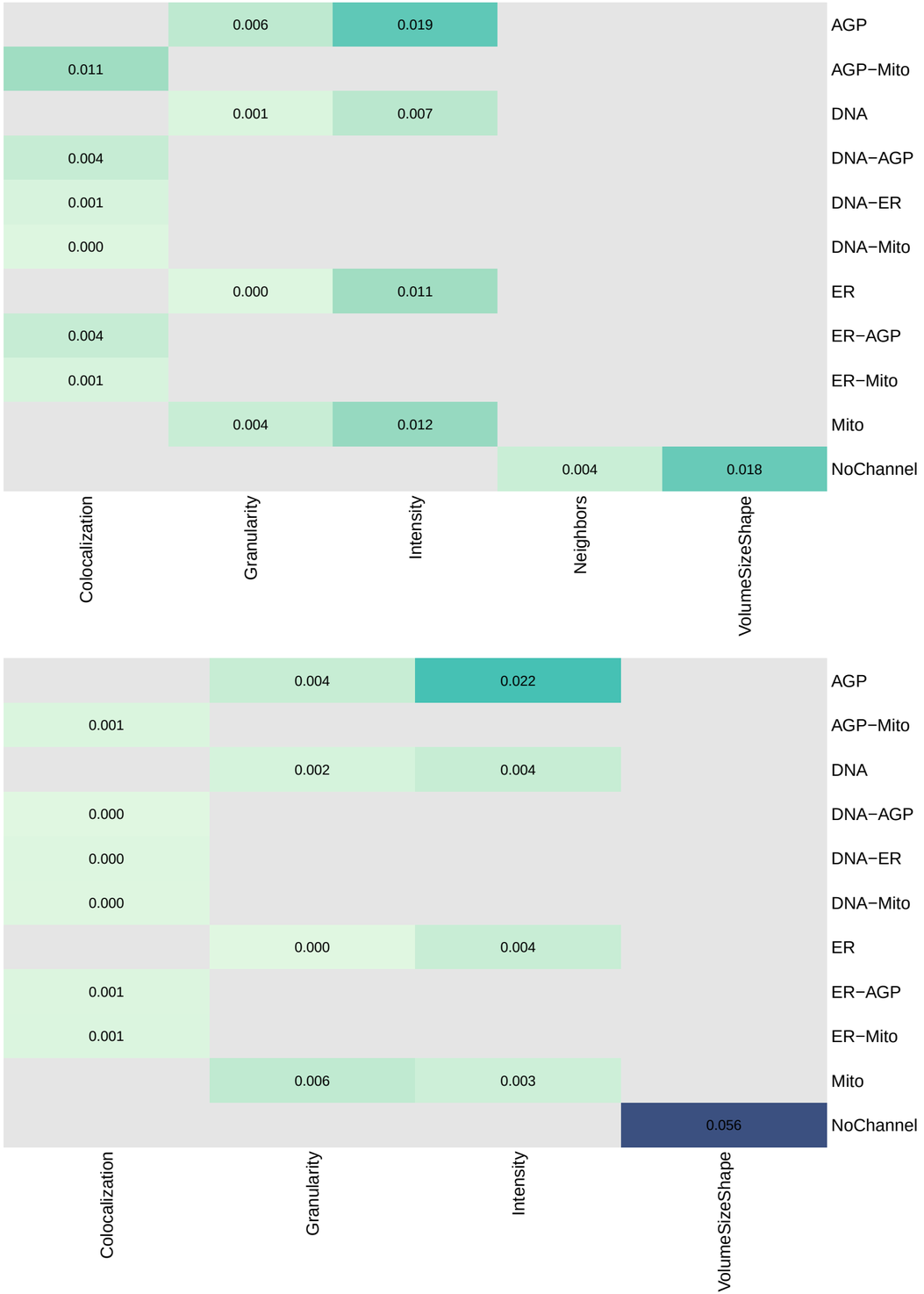
Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM

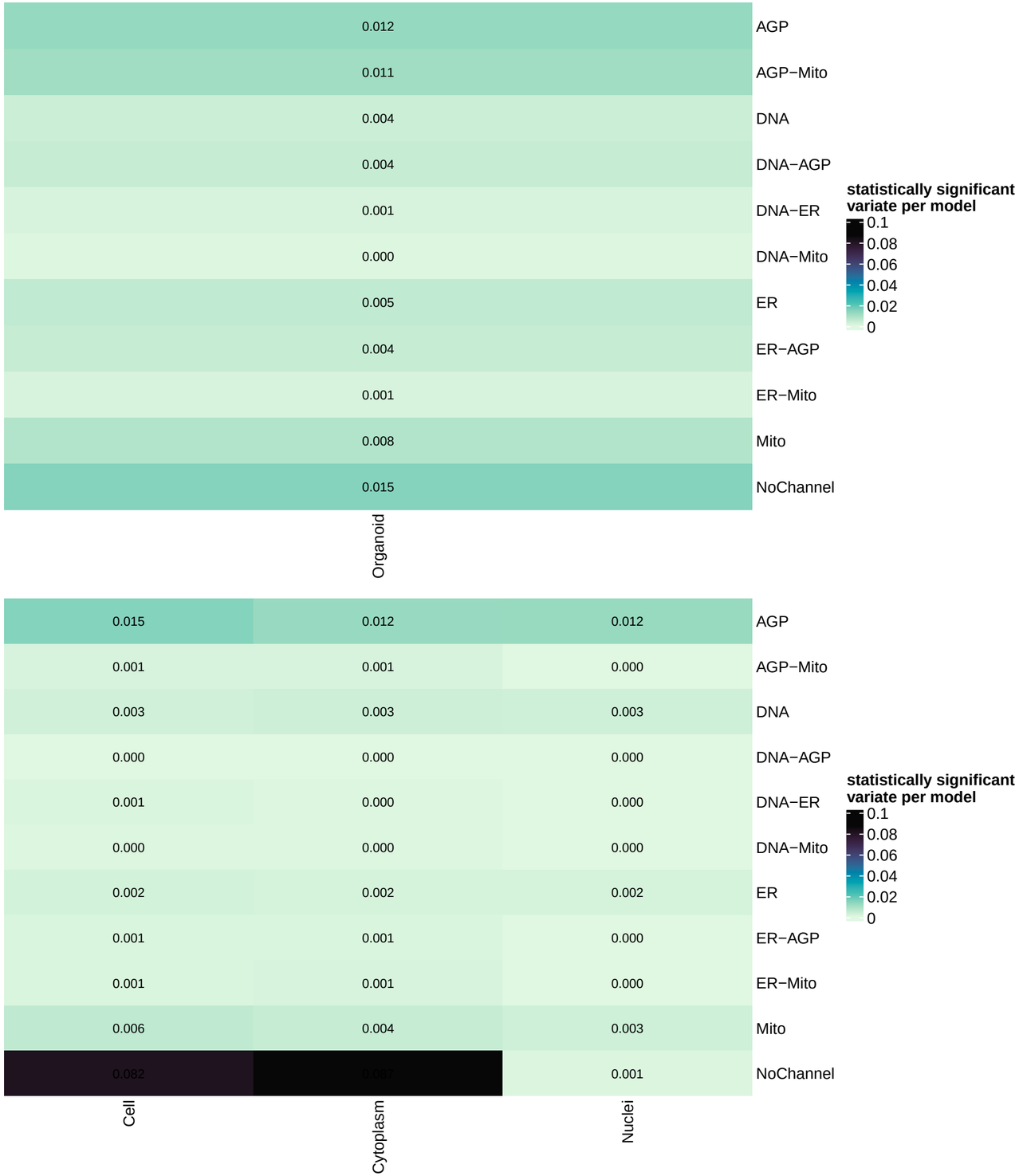
statistically significant variate per model

0.1
0.05
0


organoid
sc
Staurosporine_10nM
Fimepinostat_1uM
Copanlisib_1uM
Everolimus_1uM
Rapamycin_1uM
Sapanisertib_1uM
Vistusertib_1uM
Imatinib_1uM
Nilotinib_1uM
Cabozantinib_1uM
Linsitinib_1uM
Panobinostat_1uM
ARV−825_1uM
Onalespib_1uM
Digoxin_1uM
Ketotifen_1uM
Trabectedin_1uM
Binimetinib_10uM
Binimetinib_1uM
Mirdametinib_10uM
Mirdametinib_1uM
Trametinib_10uM
Trametinib_1uM
Selumetinib_10uM
Selumetinib_1uM
−20
−10
0
10
20
−25
0
25
50
number of significant variates (down < 0 < up)
direction
down
up

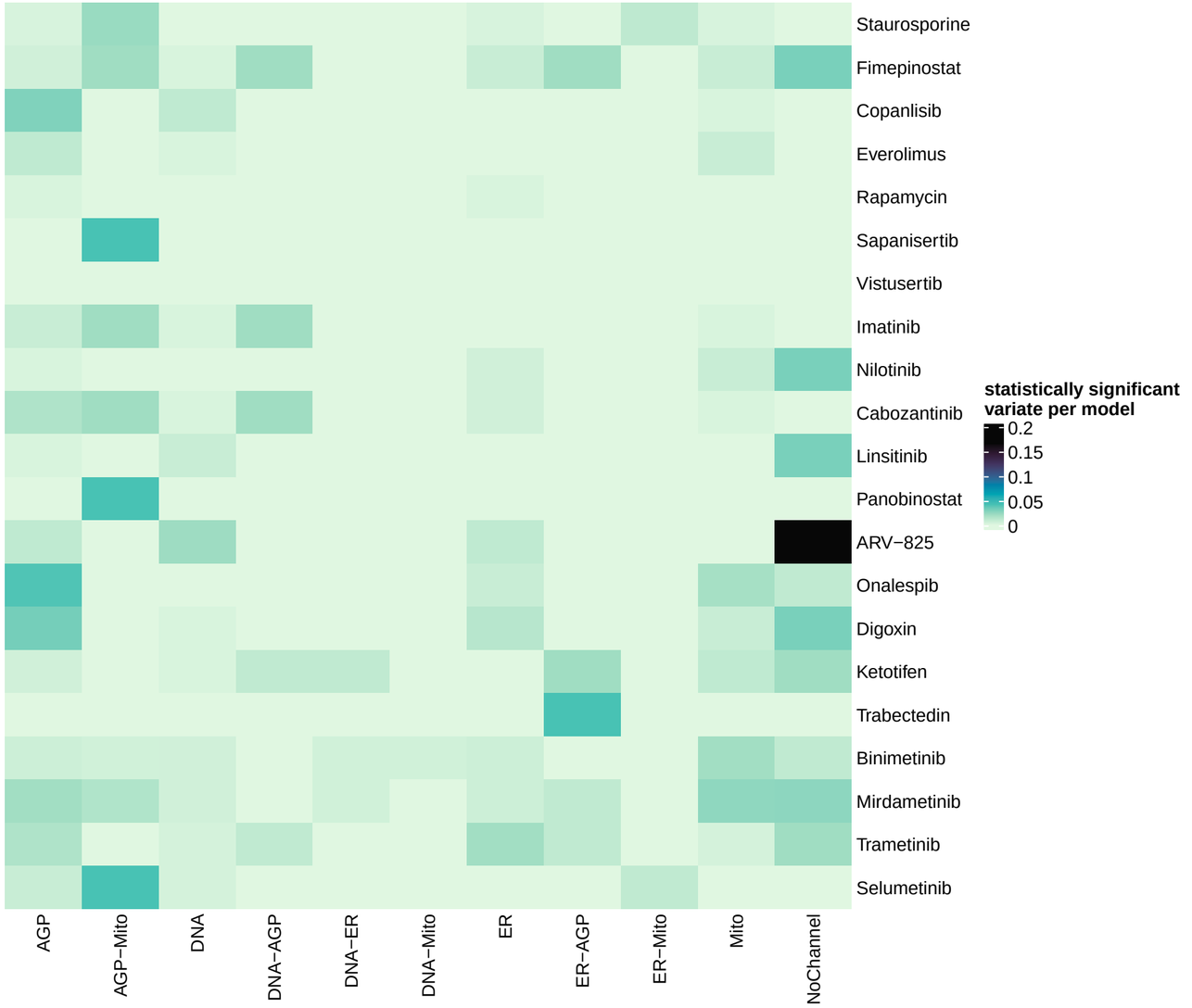

compartment_label
channel_label
Feature_type
organoid
sc
Intensity
VolumeSizeShape
Neighbors
Colocalization
Granularity
NoChannel
AGP
AGP−Mito
Mito
ER
ER−AGP
DNA−AGP
DNA
ER−Mito
DNA−ER
DNA−Mito
Organoid
VolumeSizeShape
Intensity
Granularity
Colocalization
DNA−AGP
NoChannel
AGP
Mito
DNA
ER
AGP−Mito
ER−Mito
ER−AGP
DNA−ER
DNA−Mito
Cell
Cytoplasm
Nuclei
−1.0
−0.5
0.0
0.5
−4
−2
0
2
4
log2 odds ratio of being significant
FDR < 0.05
depleted
enriched
n.s.
NA

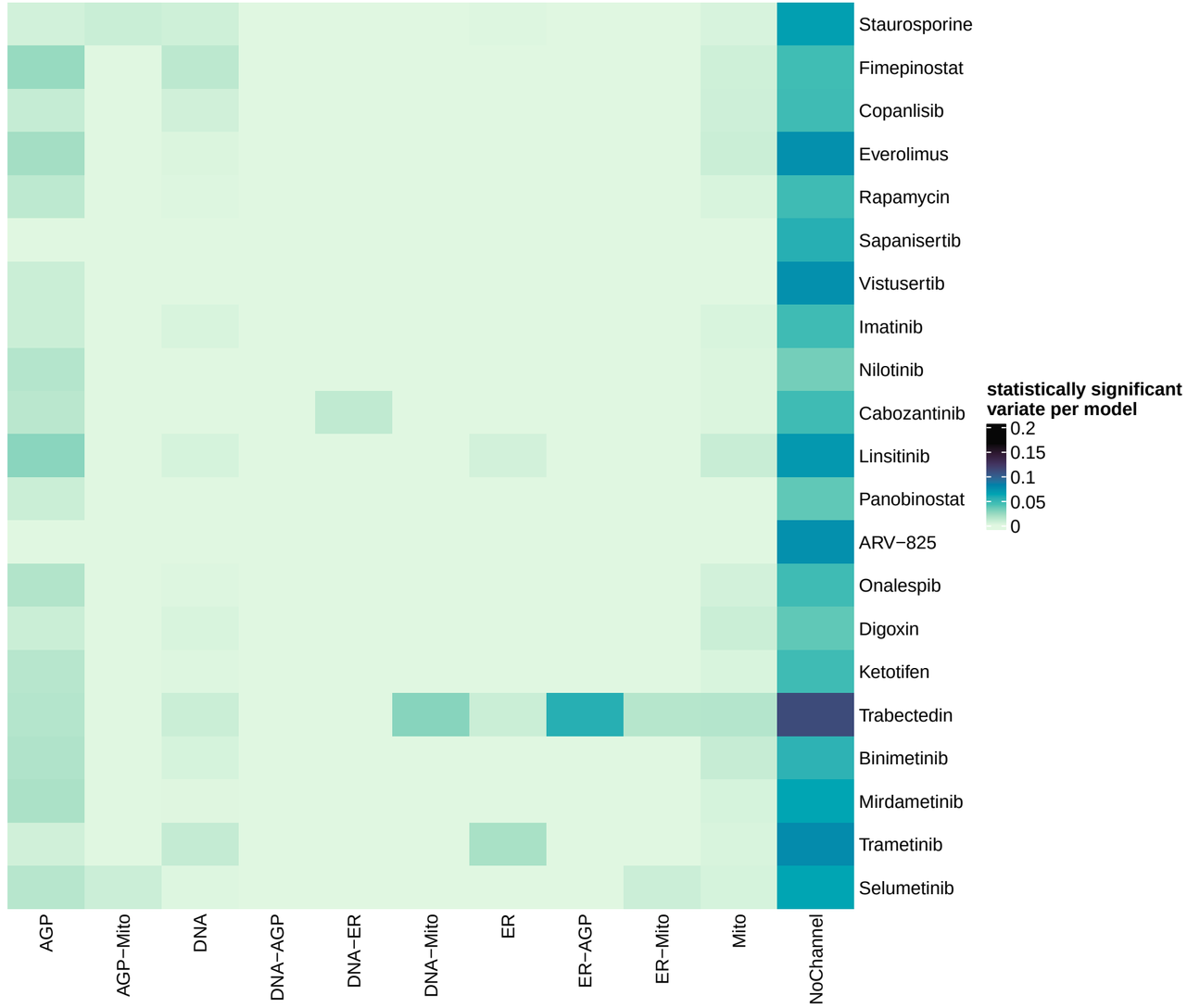

AGP
AGP−Mito
DNA
DNA−AGP
DNA−ER
DNA−Mito
ER
ER−AGP
ER−Mito
Mito
NoChannel
0.006
0.019
0.011
0.001
0.007
0.004
0.001
0.000
0.000
0.011
0.004
0.001
0.004
0.012
0.004
0.018
Colocalization
Granularity
Intensity
Neighbors
VolumeSizeShape
statistically significant variate per model
0.1
0.08
0.06
0.04
0.02
0


AGP
AGP−Mito
DNA
DNA−AGP
DNA−ER
DNA−Mito
ER
ER−AGP
ER−Mito
Mito
NoChannel
0.012
0.011
0.004
0.004
0.001
0.000
0.005
0.004
0.001
0.008
0.015
Organoid
statistically significant variate per model
0.1
0.08
0.06
0.04
0.02
0

AGP
AGP−Mito
DNA
DNA−AGP
DNA−ER
DNA−Mito
ER
ER−AGP
ER−Mito
Mito
NoChannel
0.004
0.022
0.001
0.002
0.004
0.000
0.000
0.000
0.004
0.001
0.001
0.006
0.003
0.056
Colocalization
Granularity
Intensity
VolumeSizeShape
statistically significant variate per model
0.1
0.08
0.06
0.04
0.02
0

AGP
AGP−Mito
DNA
DNA−AGP
DNA−ER
DNA−Mito
ER
ER−AGP
ER−Mito
Mito
NoChannel
0.015
0.012
0.012
0.001
0.001
0.000
0.003
0.003
0.003
0.000
0.000
0.000
0.001
0.000
0.000
0.000
0.000
0.000
0.002
0.002
0.002
0.001
0.001
0.000
0.001
0.001
0.000
0.006
0.004
0.003
0.082
0.087
0.001
Cell
Cytoplasm
Nuclei
statistically significant variate per model
0.1
0.08
0.06
0.04
0.02
0


Staurosporine
Fimepinostat
Copanlisib
Everolimus
Rapamycin
Sapanisertib
Vistusertib
Imatinib
Nilotinib
Cabozantinib
Linsitinib
Panobinostat
ARV−825
Onalespib
Digoxin
Ketotifen
Trabectedin
Binimetinib
Mirdametinib
Trametinib
Selumetinib
AGP
AGP−Mito
DNA
DNA−AGP
DNA−ER
DNA−Mito
ER
ER−AGP
ER−Mito
Mito
NoChannel
statistically significant variate per model
0.2
0.15
0.1
0.05
0


Staurosporine
Fimepinostat
Copanlisib
Everolimus
Rapamycin
Sapanisertib
Vistusertib
Imatinib
Nilotinib
Cabozantinib
Linsitinib
Panobinostat
ARV−825
Onalespib
Digoxin
Ketotifen
Trabectedin
Binimetinib
Mirdametinib
Trametinib
Selumetinib
AGP
AGP−Mito
DNA
DNA−AGP
DNA−ER
DNA−Mito
ER
ER−AGP
ER−Mito
Mito
NoChannel
statistically significant variate per model
0.2
0.15
0.1
0.05
0

	Binimetinib_1.0uM	Binimetinib_10.0uM	Mirdametinib_1.0uM	Mirdametinib_10.0uM	Trametinib_1.0uM	Trametinib_10.0uM	Selumetinib_1.0uM	Selumetinib_10.0uM
Binimetinib_1.0uM	1.00	0.12	0.20	0.29	0.26	0.15	0.15	0.11
Binimetinib_10.0uM	0.12	1.00	0.27	0.28	0.31	0.12	0.13	0.10
Mirdametinib_1.0uM	0.20	0.27	1.00	0.19	0.16	0.25	0.21	0.12
Mirdametinib_10.0uM	0.29	0.28	0.19	1.00	0.36	0.15	0.07	0.16
Trametinib_1.0uM	0.26	0.31	0.16	0.36	1.00	0.36	0.26	0.21
Trametinib_10.0uM	0.15	0.12	0.25	0.15	0.36	1.00	0.37	0.31
Selumetinib_1.0uM	0.15	0.13	0.21	0.07	0.26	0.37	1.00	0.41
Selumetinib_10.0uM	0.11	0.10	0.12	0.16	0.21	0.31	0.41	1.00

Spearman: 1, 0.5, 0, −0.5, −1

	Binimetinib_1.0uM	Binimetinib_10.0uM	Mirdametinib_1.0uM	Mirdametinib_10.0uM	Trametinib_1.0uM	Trametinib_10.0uM	Selumetinib_1.0uM	Selumetinib_10.0uM
Binimetinib_1.0uM	1.00	0.38	0.38	0.16	0.26	0.27	0.31	0.20
Binimetinib_10.0uM	0.38	1.00	0.43	0.24	0.30	0.27	0.21	0.11
Mirdametinib_1.0uM	0.38	0.43	1.00	0.16	0.25	0.28	0.32	0.31
Mirdametinib_10.0uM	0.16	0.24	0.16	1.00	0.20	0.34	0.13	0.14
Trametinib_1.0uM	0.26	0.30	0.25	0.20	1.00	0.35	0.30	0.17
Trametinib_10.0uM	0.27	0.27	0.28	0.34	0.35	1.00	0.39	0.31
Selumetinib_1.0uM	0.31	0.21	0.32	0.13	0.30	0.39	1.00	0.44
Selumetinib_10.0uM	0.20	0.11	0.31	0.14	0.17	0.31	0.44	1.00

Spearman: 1, 0.5, 0, −0.5, −1

	Binimetinib_1.0uM	Binimetinib_10.0uM	Mirdametinib_1.0uM	Mirdametinib_10.0uM	Trametinib_1.0uM	Trametinib_10.0uM	Selumetinib_1.0uM	Selumetinib_10.0uM
Binimetinib_1.0uM	1.00	0.21	0.27	0.21	0.36	0.28	0.08	0.28
Binimetinib_10.0uM	0.21	1.00	0.16	0.18	0.31	0.34	0.23	0.31
Mirdametinib_1.0uM	0.27	0.16	1.00	0.18	0.25	0.24	0.17	0.12
Mirdametinib_10.0uM	0.21	0.18	0.18	1.00	0.24	0.19	0.17	0.28
Trametinib_1.0uM	0.36	0.31	0.25	0.24	1.00	0.46	0.24	0.26
Trametinib_10.0uM	0.28	0.34	0.24	0.19	0.46	1.00	0.31	0.23
Selumetinib_1.0uM	0.08	0.23	0.17	0.17	0.24	0.31	1.00	0.33
Selumetinib_10.0uM	0.28	0.31	0.12	0.28	0.26	0.23	0.33	1.00

Spearman: 1, 0.5, 0, −0.5, −1

	Binimetinib_1.0uM	Binimetinib_10.0uM	Mirdametinib_1.0uM	Mirdametinib_10.0uM	Trametinib_1.0uM	Trametinib_10.0uM	Selumetinib_1.0uM	Selumetinib_10.0uM
Binimetinib_1.0uM	1.00	0.21	0.37	0.18	0.24	0.28	0.28	0.25
Binimetinib_10.0uM	0.21	1.00	0.34	0.33	0.22	0.38	0.14	0.12
Mirdametinib_1.0uM	0.37	0.34	1.00	0.23	0.21	0.24	0.36	0.37
Mirdametinib_10.0uM	0.18	0.33	0.23	1.00	0.19	0.29	0.14	0.13
Trametinib_1.0uM	0.24	0.22	0.21	0.19	1.00	0.43	0.25	0.21
Trametinib_10.0uM	0.28	0.38	0.24	0.29	0.43	1.00	0.32	0.23
Selumetinib_1.0uM	0.28	0.14	0.36	0.14	0.25	0.32	1.00	0.54
Selumetinib_10.0uM	0.25	0.12	0.37	0.13	0.21	0.23	0.54	1.00

Spearman: 1, 0.5, 0, −0.5, −1